

The Effects of Font Disfluency on Information Retention and Reading Speed of High Schools  
Students in Online Learning

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### Abstract

In an academic environment that has increasingly moved towards using digital materials, it has never been more important to study the effects of the many aspects involved in learning virtually. One aspect, font disfluency, shows to have a potential effect on how we retain information, and with research done, both found by previous researchers and this research paper, educators can improve how students learn by just changing a font. This paper serves to answer the question: how do the effects of font disfluency, the readability of certain typefaces affect the abilities to retain information and read quickly for students at the Academic Magnet High School? The fonts that were analyzed in this study were Times New Roman, a fluent font, *Corsiva*, an easy-to-read disfluent font, and *Aquaflina*, a hard-to-read disfluent font. These typefaces were chosen to discover how fonts of increasing disfluency can affect the reading retention and reading speed of high-school students. In the data collection, students were asked to complete a short fictional reading passage given as much time as they needed. Passages were given in one of the three fonts listed above. The time students took to read the passage was recorded to analyze reading speed. After reading, students were asked to complete a short questionnaire to distract them from the information presented in the passage. Lastly, students completed a 15-question assessment which evaluated basic information in the passage, such as setting, character names, and plot events. After data was collected, evidence was found that there were slight differences between the scores each font induced. *Corsiva* fonts resulted in slightly better scores than Times New Roman and *Aquaflina* respectively. When analyzing why a more disfluent font *Aquaflina*, resulted in a worse testing score, it is probable that the difficulty in trying to read the passage could have actually harmed the student's retention. When analyzing reading speed, as font disfluency

increased, the time spent reading increased as well. This suggests that the difficulty of trying to read the words caused students to spend more time trying to read it overall.

### Introduction

There are a plethora of different ways educators can change classroom materials to improve learning. One of those aspects that often go overlooked, is that of fonts. By changing how fonts are designed, could there be a possible effect on learning? Researchers in the discipline of typography, the field of fonts, have already found possible boosts in the learning potential of students just by changing aspects of fonts (Diemand-Yauman 2). One particular feature of fonts, that of font disfluency, the readability of a certain typeface, has demonstrated a large potential for possible supplementation in information retention. If just changing a font can influence how we learn, further research should be conducted to reveal possible boosts to our abilities to retain information. This study in particular analyzes three fonts in specific with increasing disfluency, Times New Roman, *Corsiva*, and *Aquaflow*, and how they affect information retention and reading speed. In researcher Jason Geller's definition of font disfluency, phrases that use cursive fonts cause "their letters [to] connect, creating potential problems in letter segmentation" (Geller 3). As the level of cursive increases, so does the disfluency of the specific font. In this study, Times New Roman was considered an easy-to-read font as it did not utilize cursive typewriting and has large spacings between the letters. *Corsiva* is defined as an easy-to-read disfluent font because some level of cursive typewriting is used, but is still legible and easy to comprehend. *Aquaflow* was chosen as a hard-to-read disfluent font because the use of cursive typewriting is far more prevalent than in the previous fonts and disrupts understanding of the text. Overall, with knowledge of why these fonts are considered disfluent, I hypothesize that

by increasing font disfluency, information retention and reading speed will increase as students will spend more effort trying to read the passage effectively. With further research on fonts, possible boosts to learning can be found to support the education industry.

### **The Effects of Fonts on Learning**

Previous research has already shown that the different elements of fonts can have distinguishable effects on one's ability to absorb and perceive information. In a published study from the Canadian Center of Science and Education, researchers Hassan Soleimani & Elham Mohammadi experimented with whether font elements, that of font type, font size, and line spacing would have any effect on reading speed or comprehension. To figure this out, the researchers conducted two concurrent experiments that analyzed varying font elements' influence on reading speed and reading retention. In the first experiment, subjects were asked to complete a reading passage in a limited time. After, participants completed an assessment involving the information presented in the passage. Each subject was randomly assigned a passage with a variable from one of the three font elements: 10pt or 12pt font, double or single-spaced lining, and serif type or sans-serif type fonts. The second experiment the researchers conducted was similar, however, subjects would be given as much time as they needed to complete the test. The researchers found that there was no significant difference in the reading speed of sans-serif and serif fonts and double-spaced and single-spaced text. However, there had been a significant difference between the reading speed of 10pt and 12pt font, with larger fonts resulting in more questions correctly answered in the given amount of time. In the second experiment, all three font elements, font size, font type, and line spacing resulted statistically insignificant in reading comprehension. Although the study was able to successfully distinguish one statistically

significant result, it was not enough to amass much attention. Further research on font elements, specifically that of the effects of font type on reading comprehension, have become an important point of discussion among researchers. A study conducted by researchers Michael Gasser and colleagues at the University of Northern Iowa examined the effects of font types on information retention. For their experiment, researchers chose commonly used fonts found on online document software. The fonts chosen were Palatino, a serif based font, and Helvetica, a sans-serif based font. Reading passages of the different fonts were given to 149 participants, all of which were undergraduates at public universities. After reading the passage, participants were given a questionnaire that asked students how they felt about the passage. Participants then were asked to complete an evaluative assessment on the information presented in the passage, testing how well participants retained the data. The researchers found that participants reading information from serif based fonts elicited a significant recall improvement than their sans-serif counterparts. As to why, the researchers concluded that the “markings that make rows of text appear to [be] set upon a line” (Gasser et al. 185) in serif fonts helps reading become easier to process, meaning that “more attentional resources can then be devoted to attending to the message in the text, which results in deeper processing and an easier recall of the information presented” (185). The research overall found that small aspects of fonts could cause great differences in perceptual performance, motivating further research into this field, specifically that of font disfluency. When analyzing just font disfluency, I hypothesize that with an increasing amount of font disfluency, there will be a resulting significant increase in learning abilities.

### **Benefits of Font Disfluency**

Font disfluency has become one of the most prominent areas of focus in typography research. The most influential study on font disfluency, conducted by Connor Diemand-Yauman and colleagues, studied the potential effects that font disfluency could have in a classroom environment. The study was performed in a public high school, across multiple classes in different academic fields, and with participation of two hundred twenty-two students. In their study, the researchers changed all the fonts of the learning materials that classes used to more disfluent fonts. Examples of learning materials included powerpoints and reading passages. Teachers would then teach the class as normal, teaching the same material for two months. To examine the effects of the study, the researchers collected evaluative assessments that students from both fluent and disfluent completed. The study found that classes that utilized disfluent fonts resulted in higher performance in assessments. The researchers concluded that disfluent fonts could be an easy and affordable way to help students when learning new material, emphasizing that “disfluency leads to deeper processing”. This study has become one of the most influential studies in the field, sparking further conversation into how font disfluency can help information learning. A similar study by Jason Geller and colleagues continued research into font disfluency and how it affects one’s ability to perceive information. The study was performed among thirty undergraduates at a public university. In the experiment, the researchers showed participants 99 different words, A third of which were presented in common type-print fonts, easy-to-read cursive fonts, and hard to read cursive fonts respectively. Participants were asked to name each word before moving to the further ones. After recognizing all the words, the participants were then asked to label as many United States capitals as they could. This distracted them from thinking about any words shown. Participants were then presented with words that either appeared during the study, in the same font, or new words in a random selection of one of

the three fonts. Participants could select if the words were either from the last experimental batch or if they were just added. Participants could take as much time as they needed during decision-making. The researchers concluded that participants recalled easy-to-read disfluent fonts better than common type-print fonts followed by hard-to-read disfluent fonts. The researchers hypothesized from their results that more disfluent fonts did not perform better because that higher levels of disfluency distracted the readers too much to effectively comprehend what was being shown.

### **Null/Negative Effects of Font Disfluency**

Although there have been studies supporting the benefits of font disfluency on learning retention, there have also been multiple studies that actually supported a negative correlation between the two. In a study conducted by psychology majors at the university of Waikato, researchers Andrea Taylor and colleagues researched the effects of Sans Forgetica, a font designed by RMIT University to replicate the effects of font disfluency and therefore effectively boost learning retention. The researchers conducted their experiment in the same way that the creators of Sans Forgetica did in the attempt to recreate the initial methodology. In the study, 156 Participants were shown twenty pairs of words in either Arial or in Sans Forgetica for a time of 100 ms per pair. The pairs shown were highly associated words “for example, accident – crash, chip – potato”. After seeing all twenty pairs, participants were then randomly shown a word from one of the twenty pairs and asked to find the second associated word. All the words in the memory tests were written in Times New Roman to avoid effects that might result from using the fonts previously used. The researchers found subjects had recalled fewer words when written in Sans Forgetica, directly the opposite result that the creators of the font had seen when creating the font. The researchers concluded that Sans Forgetica actually impaired people’s ability to

recall information implying that further and more specific research needs to be conducted to accurately demonstrate the effects of font disfluency. In a similar study on the effectiveness of font disfluency on learning retention, a meta-analysis by Andrew Meyer at Yale University and colleagues across various American universities explored the effects of font disfluency on answering counterintuitive math problems. In their research, Meyer and his colleagues examined previous research on font disfluency's effects on perception and used it to construct a study that would test any link between font disfluency and analytical reasoning of math ability across a number of universities. The experiment involved utilizing font disfluencies in math passages to see if it would have any effect on correctly answering counterintuitive math problems. The study evaluated seventeen American universities and concluded that there was a null or even negative effect on solving math problems when disfluent fonts were used. However, disfluent fonts did increase the speed at which the questions were answered. The researchers concluded that the use of disfluent fonts fails to benefit reasoning, however still suggesting that disfluency may benefit memory. Although the general research of font disfluency is inconsistent, further and more specific research will lead to more conclusive data to be found.

### **Identifying the Gap**

With inconsistent data on the effects of font disfluency on information retention, it is necessary for further research to be done to effectively conclude any significant results. As well, with an increasing switch to digital learning in the classroom environment, it has become increasingly important to evaluate the effects of font disfluency on digital learning, as most studies have been conducted on paper. With the minimal amount of research done on the effects of processing disfluent fonts from a digital screen instead of on paper, I will add on to the general

frame of knowledge in the field and add needed information to an unfilled gap. Based on the ideas of Jason Geller's paper, I will study the effects of three fonts with increasingly more disfluent natures, those of Times New Roman, Corsiva, and *Aquafina* respectively. Overall, this study researches the effects of various levels of disfluency on digital learning to effectively find a way typography can benefit student's learning performance in an increasingly more digital environment.

## **Methodology**

### **Overview**

The methodology used in this research is correlative with many of the studies focusing on the field of font disfluency. Implementing reading passages with different levels of font disfluency has shown that there could be significant results when considering information retention and reading speed. Another important factor I have decided to include was adding a questionnaire, which Jason Geller used to distract students from the information in the experiment.

Administering a questionnaire based on students' opinions on topics about pizza allows the participant to have their focus taken away from the information presented in the passage employing the assessment to just be based on information retention. Lastly, utilizing a 15-question assessment based on the information in the passage allows for the effective evaluation of reading retention based on font disfluency. The study was conducted on a digital medium to replicate the growing usage of digital schooling in classrooms.

### **Subject Demographics and Medium of Examination**

Subjects chosen for this study were all high school seniors in AP english classes at the Academic Magnet High School. Students in AP classes were specifically chosen to analyze students amongst similar reading skill levels. As well, high school students were chosen for this study because the effects of font disfluency on information retention and reading speed in classroom environments would be highly valuable as finding ways to learn effectively and efficiently can be used to bolster grades and academic success. The study was done in a duration of a month with trials occurring every other day. In examination of the medium in which the reading passage

was implemented, students read from a digital screen to mimic the increasing use of technology in the classroom. Especially as many schools become virtual, it has become increasingly important to analyze the effects of learning from a digital medium as learning digitally has become the new norm.

### Fonts Used

The fonts used in the passage are in increasing disfluency as shown below.

Fluent - Times New Roman

Easy-to-read disfluent - *Corsiva Script*

Hard-to-read disfluent - *Aquaflina Script*

When used in an example sentence: The beautiful moon shined bright

Fluent - The beautiful moon shined bright.

Easy-to-read Disfluent - *The beautiful moon shined bright.*

Hard-to-read disfluent - *The beautiful moon shined bright.*

### Reading Passages

Participants were shown one of three digital reading passages, each with increasing levels of font disfluency. All passages were double-spaced to mimic reading passages that students often see online. In addition, all passages were presented in 12 pt font as it has become standard to present readings in this size font. To avoid any chance that the reader would be familiar with the information in the passage, an original fictitious piece was developed solely for the experiment. In Passage 1 (Appendix A) commonly used fluent font Times New Roman was used. In passage 2 (Appendix B) disfluent font *Corsiva Script* was used. In passage 3 (Appendix C) extremely

disfluent *Aquafina* Script was used. The times students took to read passages was recorded to illustrate the effects of reading speed of increasingly disfluent fonts. Students were told to read the passage once in the attempt to prevent times of reading from being too long.

### **Questionnaire**

After students completed the reading, they were given a short questionnaire (Appendix D) based on aspects of pizza. The questions were irrelevant to the information in the reading passage and the answers were not recorded as they were insignificant in the scope of the study. The purpose of the questionnaire was to keep students' focus off the information in the passage in the attempt to solely expose the effects of font disfluency on information retention.

### **Assessment**

After completing the Questionnaire, a 15-question assessment (Appendix E) was given to participants. Questions were based on information presented in the passage such as actions in the plot, character names, and locations in the setting. These questions focused on the student's ability to recall basic information presented in the text rather than a student's ability to interpret or analyze information as the answers can be found directly in the passage. By testing one's ability to recall only basic information found in the text, any outside variable of interpretation or analytical skills will be eliminated. The assessment is administered digitally to mimic online tests that many teachers are now implementing in classroom settings. As well, a single font, Arial, was used for the assessment to eliminate any effect that font disfluency might have when reading questions. After the assessment was completed, answers were recorded and scored for correctness (Appendix F). Scores were then linked to their corresponding font type and examined

for any correlation between font disfluency and correctly answering test questions/information retention.

### **Complete Process of Experimentation**

Once students arrived for testing, the digital reading passage was shown and students were given unlimited time to read the passage. Once the students acknowledged that they had finished reading, the researcher recorded the time they took in order to examine reading speed. After completion of the passage, students were given an on-paper questionnaire as a way to distract them from thinking of the information in the passage. Once the questionnaire was completed, students were given a digital 15-question assessment based on the information in the passage. Students were given unlimited time to complete the assessment. Once finished, assessments were submitted and analyzed for the number of questions correctly answered. The number of answers a student correctly answered was counted towards their level of font disfluency and examined for correlatory effects between font disfluency and information retention.

### Results

| Times New Roman | <i>Corsiva</i> | <i>Aquaфина</i> |
|-----------------|----------------|-----------------|
| 13              | 13             | 8               |
| 9               | 7              | 10              |
| 9               | 11             | 10              |
| 10              | 11             | 8               |
| 12              | 13             | 10              |

Figure 1: Table of answers correct for each font (out of 15)

| Times New Roman | <i>Corsiva</i> | <i>Aquaфина</i> |
|-----------------|----------------|-----------------|
| 10.6            | 11             | 9.2             |

Figure 2: Table of averages of correct answers for each font (out of 15)

| Times New Roman | <i>Corsiva</i> | <i>Aquaфина</i> |
|-----------------|----------------|-----------------|
| 115             | 159            | 192             |
| 192             | 190            | 197             |
| 190             | 180            | 312             |
| 115             | 175            | 237             |
| 141             | 140            | 192             |

Figure 3: Table of time spent reading (in seconds) for each font

| Times New Roman | <i>Corsiva</i> | <i>Aquaфина</i> |
|-----------------|----------------|-----------------|
| 150.6           | 168.8          | 226             |

Figure 4: Table of averages for time spent reading (in seconds) each font

## Data Analysis

As classrooms further implement technology in the classroom, research on learning through digital mediums become increasingly important to maximize students' potential learning ability. One important aspect, those of the types of fonts used, has become an important focus of research in the field. Studies like those conducted by Jason Geller determined that easy-to-read disfluent fonts offered higher levels of reading retention while other studies, like a metaanalysis conducted by Andrew Meyer, found out that disfluent fonts worsened reading retention in comparison to more fluent fonts. In a field of research that has conflicting results on the supposed effects of fonts, this research paper delves into three specific fonts, Times New Roman, *Corsiva* and *Aquaflow* and how through a digital medium these affect reading speed and reading retention. The reason these three fonts were chosen was because they represented varying levels of font disfluency. Times New Roman was chosen as it is a typeface commonly chosen in font disfluency studies as a representative example of fluent fonts. Without a distinguishable use of cursive typewriting, Times New Roman is simple and easy to comprehend. *Corsiva* was chosen as an easy-to-read disfluent font as it utilizes some levels of cursive typewriting, however still is clear and readable. *Aquaflow* was chosen as a hard-to-read font as its extensive use of cursive typewriting causes trouble for readers to correctly identify words (Geller 3). Furthermore, in this paper's trials, students reading passages using *Aquaflow* fonts conveyed difficulties understanding sentences of the text, emphasizing its status as hard to read. With little research done on these three fonts in specific, along with a lack of research on digital mediums, the study fills a missing gap in the knowledge base.

Upon analyzing the data, the study finds that font disfluency has minimal effects on information retention for students at the Academic Magnet High School. Students who read passages in fluent font Times New Roman answered correctly on average 12 questions, students who read from easy-to-read disfluent font *Corsiva* answered correctly on average 13 questions, and students who read from hard-to-read disfluent font *Aquafina* answered correctly on average 10 questions. Comparing the results between Times New Roman and *Corsiva*, it is shown that the number of questions correctly answered and therefore the ability to retain information increases slightly when using easy-to-read disfluent fonts over fluent fonts. Although the difference in the number of questions correctly answered were not great enough to be considered statistically significant, the data provides insight into how easy-to-read disfluent fonts can be used to render similar, or slightly greater reading retention in the classroom. When analyzing the results of hard-to-read *Aquafina*, a drop in the average number of questions correctly answered is noticeable when compared to Times New Roman (1 answer correctly answered higher) and *Corsiva* (2 answers correctly answered higher). Although the differences between scores were too small to have statistical significance, the results suggest that when font disfluency reaches a certain point where it becomes too difficult to read, it will have negative effects on the retention of information. The effects of font disfluency on reading retention, although minimal, gives insight into how the readability of fonts can affect how students learn.

Analyzing the results of font disfluency's effects on reading speed, it is evident that how students are able to read passages can greatly affect the pace at which they read. Students who read passages using Times New Roman completed the reading 150.6 seconds. Students reading passages using easy-to-read disfluent *Corsiva* read on average in 168.8 seconds. Students reading

passages using hard-to-read disfluent *Aquafina* completed the passage on average in 226 seconds.

Evidently, as the level of font disfluency increased, the amount of time needed to complete the passage rose as well. Acknowledging the difference in time spent reading between hard-to-read *Aquafina* passages and passages with lesser levels of font disfluency, a large gap in the amount of time reading is evident. *Aquafina*, taking on average 226 seconds to complete, was 57.2 and 75.4 seconds on average longer than *Corsiva* and Times New Roman respectively. This suggests that as the level of font disfluency reaches a certain point of difficulty to read, it can cause students to stumble through words and require multiple readings of sentences to fully comprehend what they are reading. This idea is correlative with Jason Geller's experiment which discovered that excessive font disfluency deteriorated information retention. When wondering whether to utilize disfluent fonts on digital classroom materials, teachers should also consider the amount of time spent reading in addition to reading retention as a significant factor of improving learning.

## **Implications**

Understanding how fonts influence how we comprehend and digest information has large implications for the field of education. When beneficial, fonts can be an easy way to influence how students learn and retain information better. In this particular study, the use of easy-to-read disfluent fonts like *Corsiva* provoke a negligible or slightly better ability to retain information when compared to fluent fonts like Times New Roman. With the appropriate research, teachers should implement more disfluent fonts like those of *Corsiva* with the expectations that they will not negatively impact the reading retention of students. By changing fonts, students will engage with materials in a new way, and this change may entice students to continue learning further if

the font allows for a better understanding of information. To emphasize this idea, students who read from easy-to-read disfluent and fluent fonts (which had the highest recall of information) typically tended to verbally express their enjoyment of reading the passage, signifying that they would probably read more in the future. Knowing how font disfluency can improve our abilities in learning is important for the future and bettering of education, additionally, understanding when font disfluency goes too far becomes a significant aspect in the process of learning.

Another implication found in the study, that being too much font disfluency, can reduce a student's ability to read and retain information. In the data, hard-to-read disfluent font *Aquafina* associated with the least amount of questions on average correctly answered and the longest amount of time spent reading. Teachers and students alike should avoid using fonts which are too difficult to read as it might provoke negative effects on a student's learning ability.

Lastly, educators should be aware of the effects of font disfluency on the time spent reading material. As shown by the research, as font disfluency increases, the time spent reading increases substantially. Teachers at the Academic Magnet High School should avoid using fonts that are too disfluent to avoid unnecessary time spent reading. Ultimately, educators should be cognizant of the fonts they choose to use to maintain a balance between reading speed and reading retention.

### **Limitations**

One limitation, particularly, the study's focus on senior students from the Academic Magnet High School limits the results of the data to students of the school. More specifically, the study would be limited to students taking AP English classes, or students with strong abilities in reading and writing. Additionally, the study is limited to fiction passages of somewhat short

length. With many classroom materials spanning pages, using a font that may benefit students in short works like the one chosen in this experiment may ultimately affect them negatively if they have to read for an extended period of time. As well, by implementing the research through a digital medium, the data only correlates to research done on virtual education. Although classrooms are moving towards using digital material, classes still use on-paper assignments. This causes difficulties extending the results found in the paper as the research focuses only on reading materials from digital mediums. Furthermore, the findings of the data can only be made about the fonts used in the research (Times New Roman, *Corsiva*, and *Aquaflow*) as other fonts may suggest other effects on learning. This limits the research of font disfluency to these three fonts. Although the data is limited, educators should experiment with other fonts to see how they affect reading retention and reading speed.

### **Conclusion**

This study focuses on font disfluency, the readability of words, and how it affects information retention and reading speed. Based on the fundamental study by Diemand-Yuaman, the significance of font disfluency on the effects on information retention sparked future research, and in particular this study, which finds that disfluent fonts may have a negligible or a slight improvement in the ability to read and retain information. Additionally, the study found that as font disfluency increased, the time spent reading increased as well. Although previous research on font disfluency's effects on learning is conflicting, researchers should continue to study the field of font disfluency to find possible supplements to one's learning ability. Overall, with increased knowledge on the aspects of fonts, educators will be able to utilize this

information to figure out how to allow students to learn at their fullest potential with such minimal effort as changing a font.

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## Appendix

### Appendix A

At a certain age, some will say at around forty years or so, there comes a time when one reflects back on their life, excited to see their accomplishments and achievements, only to find a trophy case with no trophy. It is at this time when middle aged people become obsessed with leaving their name inscribed on a giant mountain with spotlights shining on it every night of every day, hoping to be remembered. There was a point to this fairly long introduction, but it's been lost to the thoughts of crisis. Nevertheless, there was a lady, around this age of remembrance and frantic thoughts, named Kathy.

She remembered when she was young, striking the edges of steel pans with wooden sticks loudly, laughing along to the metallic ringing it echoed. The harmony created would permeate throughout the giant house, manor Casablanca, a manor with striking resemblance to the Biltmore estate or another costly mansion. Her mother, Julie, had quickly picked her up, distancing Kathy from the percussion instrument; however her mother already realized what was about to come as she heard forceful feet two flights of stairs above, slowly descending in a dramatic fashion.

A stylish top hat with a stripe sat atop the man's head at the stair's bottom, his eyebrows furrowed with either rage or confusion. He privately took Kathy's mother aside by motioning to her with his golden-tipped cane. He spoke in a hushed voice, but frankly he was quite bad at whispering.

"Sir, I know-" Kathy's mother started explaining, but she was cut off by furious whispering.

"What I know is that my guests upstairs are trying to enjoy and relax to my exquisite piano playing, but they're being interrupted by insidious banging," he lectured sternly. "You are my servants, not some children I'm required to babysit. Take control of that tone-deaf child before I do."

His whispering grew louder and more into a father talking to an irresponsible teenager. He finished his last vowel with emphasis, before turning away and quickly jumping up the steps to the above floors.

"Come on let's go outside," Kathy's mother sighed, ushering her out of a swinging glass door into the garden, which was one of Kathy's favorite places. At the time, she didn't exactly understand the reliance she and her mother had on Mr. Dawson, the owner of the manor of which they were servants in as a result of debt owed to the man.

"He's so mean," Kathy remarked, staring at a bee buzzing through a flower bed.

"He's a very generous man," her mother responded, grabbing a watering can and beginning to shower the beds with water.

Kathy shrugged and walked over to a crowded bed with bursts of colored daisies, a sore thumb from the rest of the uniformed garden. She heard voices coming from the third story balcony where Mr. Dawson and his guests were talking eloquently when he sat down at a stool in front of a mahogany piano with ivory keys.

There was one thing Kathy didn't hate about Mr. Dawson; his seemingly prodigal piano playing which struck a chord in her heart. He had once pulled up a chair next to the Baroque

piano and instructed Kathy to sit. She carefully examined him as his eyes closed, and began slowly tapping his toes instinctually to the allegro tuned beat. Eventually, about halfway through the mind-controlling piece, his back loosened from its soldier poise and subconsciously swayed to the beat of the piano. The notes flowed through Kathy's ears like a flashing river with raging waters, until one note stuck out. Mr. Dawson crinkled his nose at the noise of a sour key and sudden tempo change, and looked at the piano disgustingly.

"Well then, run along now," he said sharply, and walked away towards the kitchen.

Kathy awoke from her trance of the memory, still staring at Mr. Dawson sitting on the edge of the stool. His fingers started to dance along the keys of the piano, his fingers like a ballerina jumping from key to key. Kathy closed her eyes and rested her fingers on the petals of a daisy. She tried to imitate Mr. Dawson's movements exactly, from crossing over her arms to reach a certain petal to remembering to curve her fingers as she struck the "keys". As her fingers ran along the petals, she hummed the mirrored melody heard from above, eventually letting her foot start to barely tap on the gravel path. She copied Mr. Dawson's keystrokes very dramatically through the crescendo, almost tearing a petal from the flower's head at the loud finale.

"Thank you," Mr. Dawson bowed to his guests. The guests started clapping impressed at the man's artistic abilities. But in Kathy's mind, they were clapping for her.

## Appendix B

*At a certain age, some will say at around forty years or so, there comes a time when one reflects back on their life, excited to see their accomplishments and achievements, only to find a trophy case with no trophy. It is at this time when middle aged people become obsessed with leaving their name inscribed on a giant mountain with spotlights shining on it every night of every day, hoping to be remembered. There was a point to this fairly long introduction, but it's been lost to the thoughts of crisis. Nevertheless, there was a lady, around this age of remembrance and frantic thoughts, named Kathy.*

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*A stylish top hat with a stripe sat atop the man's head at the stair's bottom, his eyebrows furrowed with either rage or confusion. He privately took Kathy's mother aside by motioning to her with his golden-tipped cane. He spoke in a hushed voice, but frankly he was quite bad at whispering.*

*"Sir, I know-" Kathy's mother started explaining, but she was cut off by furious whispering.*

*"What I know is that my guests upstairs are trying to enjoy and relax to my exquisite piano playing, but they're being interrupted by insidious banging," he lectured sternly. "You are my servants, not some children I'm required to babysit. Take control of that tone-deaf child before I do."*

*His whispering grew louder and more into a father talking to an irresponsible teenager. He finished his last vowel with emphasis, before turning away and quickly jumping up the steps to the above floors.*

*"Come on let's go outside," Kathy's mother sighed, ushering her out of a swinging glass door into the garden, which was one of Kathy's favorite places. At the time, she didn't exactly understand the reliance she and her mother had on Mr. Dawson, the owner of the manor of which they were servants in as a result of debt owed to the man.*

*"He's so mean," Kathy remarked, staring at a bee buzzing through a flower bed.*

*"He's a very generous man," her mother responded, grabbing a watering can and beginning to shower the beds with water.*

*Kathy shrugged and walked over to a crowded bed with bursts of colored daisies, a sore thumb from the rest of the uniformed garden. She heard voices coming from the third story balcony where Mr. Dawson and his guests were talking eloquently when he sat down at a stool in front of a mahogany piano with ivory keys.*

*There was one thing Kathy didn't hate about Mr. Dawson; his seemingly prodigal piano playing which struck a chord in her heart. He had once pulled up a chair next to the Baroque piano and instructed Kathy to sit. She carefully examined him as his eyes closed, and began slowly tapping his toes instinctually to the allegro tuned beat. Eventually, about halfway through the mind-controlling piece, his back loosened from its soldier poise and subconsciously swayed to the beat of the piano. The notes flowed through Kathy's ears like a flashing river with raging waters, until one note stuck out. Mr. Dawson crinkled his nose at the noise of a sour key and sudden tempo change, and looked at the piano disgustingly.*

*"Well then, run along now," he said sharply, and walked away towards the kitchen.*

*Kathy awoke from her trance of the memory, still staring at Mr. Dawson sitting on the edge of the stool. His fingers started to dance along the keys of the piano, his fingers like a ballerina jumping from key to key. Kathy closed her eyes and rested her fingers on the petals of a daisy. She tried to imitate Mr. Dawson's movements exactly, from crossing over her arms to reach a certain petal to remembering to curve her fingers as she struck the "keys". As her fingers ran along the petals, she hummed the mirrored melody heard from above,*

*eventually letting her foot start to barely tap on the gravel path. She copied Mr. Dawson's keystrokes very dramatically through the crescendo, almost tearing a petal from the flower's head at the loud finale.*

*"Thank you," Mr. Dawson bowed to his guests. The guests started clapping impressed at the man's artistic abilities. But in Kathy's mind, they were clapping for her.*

## Appendix C

*At a certain age, some will say at around forty years or so, there comes a time when one reflects back on their life, excited to see their accomplishments and achievements, only to find a trophy case with no trophy. It is at this time when middle aged people become obsessed with leaving their name inscribed on a giant mountain with spotlights shining on it every night of every day, hoping to be remembered. There was a point to this fairly long introduction, but it's been lost to the thoughts of crisis. Nevertheless, there was a lady, around this age of remembrance and frantic thoughts, named Kathy.*

*She remembered when she was young, striking the edges of steel pans with wooden sticks loudly, laughing along to the metallic ringing it echoed. The harmony created would permeate throughout the giant house, manor Casablanca, a manor with striking resemblance to the Billmore estate or another costly mansion. Her mother, Julie, had quickly picked her up, distancing Kathy from the percussion instrument; however her mother already realized what was about to come as she heard forceful feet two flights of stairs above, slowly descending in a dramatic fashion.*

*A stylish top hat with a stripe sat atop the man's head at the stair's bottom, his eyebrows furrowed with either rage or confusion. He privately took Kathy's mother aside by motioning to her with his golden-tipped cane. He spoke in a hushed voice, but frankly he was quite bad at whispering.*

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*"Thank you," Mr. Dawson bowed to his guests. The guests started clapping impressed at the man's artistic abilities. But in Kathy's mind, they were clapping for her.*

## Appendix D

### Pizza Questionnaire For Students at the Academic Magnet High School



Favourite Toppings(other than cheese)

.....

Stuffed crust?                      Yes/No

Pizza Shape?                      Square/Circle

Size?                                  Deep Dish/Deep Pan/Thin

Crust?                                  Crust/No Crust

Base?                                  Bread base/Dough Base/Wholemeal

How often do you eat pizza?      Once a day/twice a week/more/less

Size?                                  Large/Extra Large/Small/Medium

Do you like extra cheese?      Yes/No

How many toppings do you normally have? 1/2/3/4/5/more

Who do you usually eat it with? Friends/Family/ Yourself

Where do you usually eat it? Home/ Friends House

Type?                                  Sweet/Savoury

**Appendix E**

1. At what age around does one begin to reflect back on one's life?
2. What do middle aged people become obsessed with?
3. What is the name of the house?
4. What are some characteristics of the house?
5. What is the name of Kathy's mother?
6. Why did the man with stylish hat motion over Kathy's mother to talk?
7. Where was one of Kathy's favorite places?
8. What is the name of the man of the owner of the manor?
9. Why do Kathy and her mother rely on the owner of the manor?
10. When outside, where does Kathy go which "sticks out like a sore thumb"?
11. What did Kathy hear when she was outside?

12. What was the one thing Kathy did not hate about the owner of the mansion?
13. Towards the end of the passage, why did the owner of the mansion become disgusted?
14. What did Kathy imitate the sounds of the piano on?
15. Describe Kathy.

**Appendix F**

- 1. Forty Years/Middle Aged**
- 2. Achievements**
- 3. Manor Casablanca**
- 4. Big, like the Biltmore**
- 5. Julie**
- 6. Kathy was making a racket.**
- 7. The garden**
- 8. Mr. Dawson**
- 9. A debt they owe**
- 10. Crowded bed of colored daisies.**
- 11. Mr. Dawson's Piano Playing**
- 12. His piano-playing ability**
- 13. Kathy made a mistake.**
- 14. Flower petals**
- 15. Innocent/Sweet**